

Eight edge cases on the A10 deterministic board

7 by 5 board; goal rows 1, 2, 3; simultaneous U/D/L/R; deterministic carrier-wins-contests resolution. Players never occupy the same square. The loser of a contested square or swap receives the ball.

Selection rule: preserve the best worst-case payoff, then maximize mean payoff over opponent actions among exact security ties. Identical-score ties use action order. The secondary criterion is a documented selection convention, not a claim that Nash equilibrium is unique. No uniform tie mix is substituted for the solver policy.

Why every probability below is 0% or 100%, never a fraction: this project's headline finding is that *every* stage game on this deterministic board has a pure equilibrium -- so a single action at 100% is the correct, expected answer here, not a simplification. Fractional probabilities only show up in the sibling document, [positions.pdf](#), which covers Littman's cases on the *random-move-order* board, where mixed equilibria are the point. If you're looking for the mixed-strategy numbers, that is the document to read; this one is about what "pure but not obvious" looks like instead: ties, dominance, and the collision rule.

Case 1 selects Down for player 0: it weakly dominates Up and Left. The four zeros against Right describe optimal continuation from four new states, not a defender committed to Right forever. These are discounted stationary values with gamma 0.9, not the undiscounted 100-turn assignment.

This board's own rule keeps every stage game pure. The other four rule variants each trade that off for genuine mixed equilibria in some states -- see [coinflip_cases.pdf](#) (a fair coin decides contests instead of the carrier -- still 0 mixed, surprisingly), [tackle_cases.pdf](#) (a 50/50 duel, 56 of 760 mixed -- the richest source), [territory_cases.pdf](#) (a dense position reward, 69 of 2380 mixed), and [slip_cases.pdf](#) (movement noise, 52 of 1200 mixed).

[Interactive explorer](#) · [Audit and reproduction details](#)

1 The swap trap

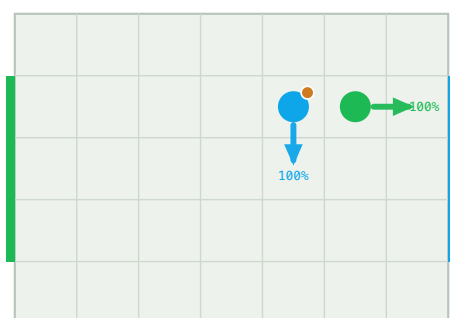
State (4, 3, 5, 3, 0) · player 0 carries the ball · $V = +0.000000$

P0 → D (100%) P1 → R (100%)

Security-tied alternatives (not probabilities): P0 U/D/L · P1 R

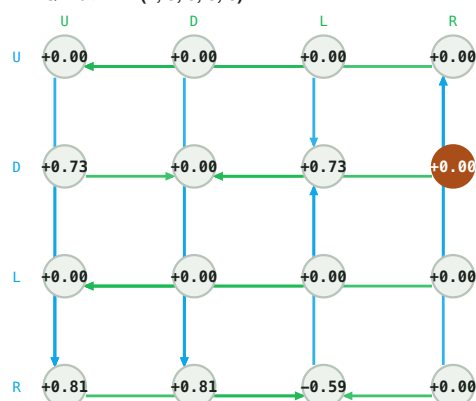
player 0's U/D/L never touch player 1 at all this turn, so all three guarantee at least 0. D weakly dominates U and L: it earns 0.729 against U/L, while U and L earn zero against every reply. The refined policy is D, not a uniform tie mix. Only R reaches player 1's square, and against L it triggers a genuine swap (verified below: (R, L) → (5,3,4,3,1)) that hands player 1 the ball, dragging R's own worst case to -0.59. Player 1's unique safe column is R for an unrelated reason: U/D/L each let a smart player 0 reach a good, independent square elsewhere in the matrix (0.81/0.81/0.729) that has nothing to do with the swap cell.

The swap trap



pure equilibrium · $V = +0.000$

Q matrix -- (4, 3, 5, 3, 0)



pure saddle at D/R

Q: PLAYER 0 PAYOFF, OPTIMAL CONTINUATION AFTER THIS TURN

P0 / P1	U	D	L	R
U	0.000000	0.000000	0.000000	0.000000
D	0.729000	0.000000	0.729000	0.000000
L	0.000000	0.000000	0.000000	0.000000
R	0.810000	0.810000	-0.590490	0.000000

ALL 16 SUCCESSOR STATES, FROM THE TRANSITION ENGINE

P0 / P1	U	D	L	R
U	(4, 4, 5, 4, 0)	(4, 4, 5, 2, 0)	(4, 4, 4, 3, 0)	(4, 4, 6, 3, 0)
D	(4, 2, 5, 4, 0)	(4, 2, 5, 2, 0)	(4, 2, 4, 3, 0)	(4, 2, 6, 3, 0)
L	(3, 3, 5, 4, 0)	(3, 3, 5, 2, 0)	(3, 3, 4, 3, 0)	(3, 3, 6, 3, 0)
R	(5, 3, 5, 4, 0)	(5, 3, 5, 2, 0)	(5, 3, 4, 3, 1)	(5, 3, 6, 3, 0)

INDEPENDENT LEARNED CROSS-CHECK: DQN VS. THE EXACT MATRIX

$\max |Q_{\text{DQN}} - Q_{\text{exact}}|$: from zero = **0.4209** · fit to exact = **0.4869**. "From zero" is 400 epochs of TD bootstrap from a random init with no access to the exact answer -- the honest baseline; "fit to exact" is direct supervised regression onto the exact matrices, isolating how well a network this size can even represent Q_{exact} before any bootstrapping noise. One representative run each (seed 0), not the multi-seed study in [docs/neural.md](#). Full per-state numbers for every board, plus a policy-gradient comparison, are in the [interactive explorer](#)'s "Show neural cross-check" panel.

P0 / P1	Exact	DQN, from zero	DQN, fit to exact
U/U	0.0000	0.2344	0.0128

U/D	0.0000	0.2672	0.1547
U/L	0.0000	0.2440	0.1559
U/R	0.0000	0.2150	0.0795
D/U	0.7290	0.3656	0.4529
D/D	0.0000	0.3172	0.1220
D/L	0.7290	0.5558	0.5696
D/R	0.0000	0.2156	-0.1531
L/U	0.0000	0.1485	-0.0693
L/D	0.0000	0.1586	-0.0122
L/L	0.0000	0.1802	0.0685
L/R	0.0000	0.1910	-0.0170
R/U	0.8100	0.4788	0.3794
R/D	0.8100	0.3891	0.3231
R/L	-0.5905	-0.2911	-0.3680
R/R	0.0000	0.2988	0.0736

2 Pinned in the corner

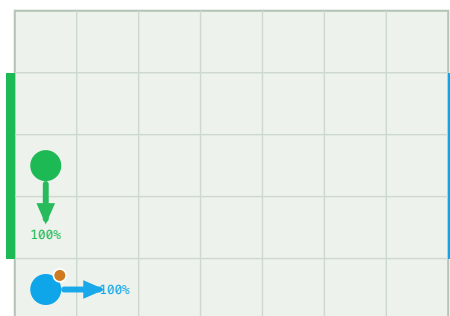
State (0, 0, 0, 2, 0) · player 0 carries the ball · $V = +0.000000$

P0 → R (100%) P1 → D (100%)

Security-tied alternatives (not probabilities): P0 D/L/R · P1 D/R

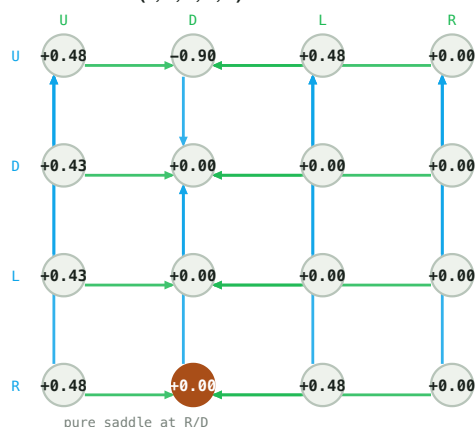
player 0's U is the one risky action: it targets the empty cell (0,1), which player 1's D also targets. Neither player is standing there, so it's an open (non-blocked) contest -- player 0 physically wins the race ((U, D) → (0,1,0,2,1)) but the A10 rule still gives the ball to the loser regardless of who reached the cell first, crashing the value to -0.9. D, L, and R never reach that cell, so they tie safely at 0.

Pinned in the corner



pure equilibrium · $V = +0.000$

Q matrix -- (0, 0, 0, 2, 0)



Q: PLAYER 0 PAYOFF, OPTIMAL CONTINUATION AFTER THIS TURN

P0 / P1	U	D	L	R
U	0.478297	-0.900000	0.478297	0.000000
D	0.430467	0.000000	0.000000	0.000000
L	0.430467	0.000000	0.000000	0.000000
R	0.478297	0.000000	0.478297	0.000000

ALL 16 SUCCESSOR STATES, FROM THE TRANSITION ENGINE

P0 / P1	U	D	L	R
U	(0, 1, 0, 3, 0)	(0, 1, 0, 2, 1)	(0, 1, 0, 2, 0)	(0, 1, 1, 2, 0)
D	(0, 0, 0, 3, 0)	(0, 0, 0, 1, 0)	(0, 0, 0, 2, 0)	(0, 0, 1, 2, 0)
L	(0, 0, 0, 3, 0)	(0, 0, 0, 1, 0)	(0, 0, 0, 2, 0)	(0, 0, 1, 2, 0)
R	(1, 0, 0, 3, 0)	(1, 0, 0, 1, 0)	(1, 0, 0, 2, 0)	(1, 0, 1, 2, 0)

INDEPENDENT LEARNED CROSS-CHECK: DQN VS. THE EXACT MATRIX

$\max |Q_{\text{DQN}} - Q_{\text{exact}}|$: from zero = **0.5168** · fit to exact = **0.6558**. "From zero" is 400 epochs of TD bootstrap from a random init with no access to the exact answer -- the honest baseline; "fit to exact" is direct supervised regression onto the exact matrices, isolating how well a network this size can even represent Q_{exact} before any bootstrapping noise. One representative run each (seed 0), not the multi-seed study in [docs/neural.md](#). Full per-state numbers for every board, plus a policy-gradient comparison, are in the [interactive explorer](#)'s "Show neural cross-check" panel.

P0 / P1	Exact	DQN, from zero	DQN, fit to exact
U/U	0.4783	0.2704	0.5110
U/D	-0.9000	-0.3832	-0.2442
U/L	0.4783	0.2191	0.3945

U/R	0.0000	0.0232	-0.0067
D/U	0.4305	0.2484	0.5116
D/D	0.0000	-0.0452	-0.0777
D/L	0.0000	0.0963	0.3703
D/R	0.0000	0.0582	-0.1063
L/U	0.4305	0.1762	0.1956
L/D	0.0000	-0.0492	-0.3925
L/L	0.0000	0.1503	0.1425
L/R	0.0000	0.0897	-0.0942
R/U	0.4783	0.3805	0.4999
R/D	0.0000	0.2344	0.2780
R/L	0.4783	0.2960	0.5072
R/R	0.0000	0.2674	0.2052

3 The standoff

State (3, 4, 4, 4, 0) · player 0 carries the ball · $V = +0.000000$

P0 → D (100%) P1 → U (100%)

Security-tied alternatives (not probabilities): P0 U/D/L · P1 U/D/R

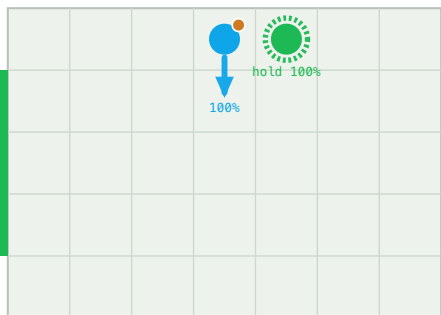
both players are already on the top row, one cell apart. U is wall-clamped (a genuine no-op) for both -- not a real move upward. R is player 0's one risky action (it reaches player 1's square: blocked against U, a real swap against L); L is player 1's mirror risk. D never touches the opponent for either player, so U/D/L (player 0) and U/D/R (player 1) all tie at the exact same safe value.

Why an all-zero matrix is correct here, not a rounding artifact: the (U, U) cell is a genuine self-loop -- `game.transitions()` confirms (U, U) → (3, 4, 4, 4, 0), the same state, with reward 0. The Bellman equation for a self-loop with zero reward is $V = 0 + 0.9 \cdot V$, whose only solution is $V = 0$ exactly for any discount below 1 -- this is algebra, not floating-point noise. The other 12 zero cells are not assumed either: every successor state they lead to -- (3,4,4,3,0), (3,4,4,4,1), (3,4,5,4,0), (3,3,4,4,0), (3,3,4,3,0), (3,3,3,4,0), (2,4,4,4,0), (2,4,4,3,0), (2,4,3,4,0), (2,4,5,4,0), (4,4,4,3,0), (4,4,3,4,1), (4,4,5,4,0) -- was looked up independently in the exact solve, and each one is itself exactly 0.0. A flat all-zero matrix is in fact the single most common shape on this board (514 of 2380 states, see case 8, "the dead zone") -- the generic outcome for a "no one has an advantage yet" configuration, not an anomaly specific to this state.

Independent learned cross-check (DQN and policy gradient): a fitted-Q DQN trained from a random initialization (400 epochs, seed 0, no access to the exact values) predicts D/L = 0.4552 and R/L = -0.2904 against the exact 0.6561 and -0.590490 -- a max error of 0.30, worse than exact but correctly recovering the sign and rough shape of every cell, and better than this project's own documented average DQN error elsewhere on this board (roughly 0.4-0.5, see docs/neural.md). DQN's own action choice from that matrix agrees with the exact solve: D for player 0 (81%), U for player 1 (86%). A self-play policy-gradient net trained the same way (1000 iterations, seed 0) does not agree here -- it converges to R for player 0 (76%) and L for player 1 (96%), exactly the exploitable swap-trap pairing this document's own case 1 warns about. That is a known limitation of on-policy self-play, not a contradiction of the exact solve: PG only trains on states its own rollouts actually reach from kickoff, and this off-the-beaten-path standoff is rarely if ever visited under the policy being trained, so nothing ever corrects it there. DQN has no such gap, because it trains against the full enumerated transition model for every state regardless of how often it is visited. Full per-state numbers for every board configuration are in the [interactive explorer](#)'s "Show neural cross-check" panel.

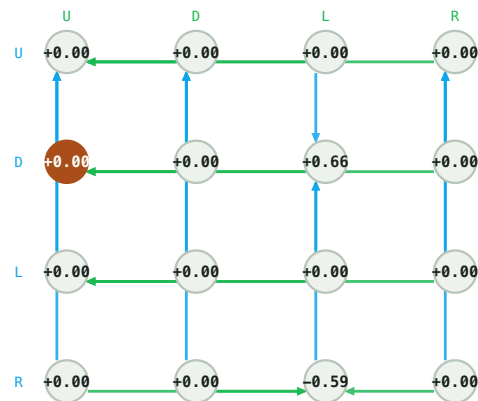
Why a differently-implemented reward would not show zeros here, and that does not make either one wrong: the field-standard convention this assignment derives from is Littman's own -- "goals are worth one point and the discount factor is set to 0.9" (Littman, *Markov games as a framework for multi-agent reinforcement learning*, ICML 1994), nothing else. That is exactly what this project implements: terminal ± 1 on a goal, 0 on every other step, $\gamma=0.9$ -- verified directly against `game.transitions()` for every rule in the A10 spec, not just this state (`tests/test_game.py` , `tests/test_a10_self_loop_zero.py`). A reward function that instead pays something on every step -- a distance shaping term, a possession bonus, anything beyond the terminal goal -- will not produce exact zeros at a state like this one, because the Bellman equation for its self-loop becomes $V = r_0 + 0.9 \cdot V$ for some $r_0 \neq 0$, not $V = 0 + 0.9 \cdot V$. This project's own `scoring="territory"` mode is a concrete, checkable example: re-solved at this exact state with a dense per-step reward added, only 5 of 16 cells stay at zero (down from 14), and the safe row carries a nonzero -0.05 floor instead. Two implementations disagreeing here is expected when their reward functions genuinely differ -- it says nothing about which one, if either, has a bug -- and only one of them is scored against the terminal ± 1 convention this assignment actually specifies. The all-zero pattern itself is not particular to one hand-picked state either: `tests/test_a10_self_loop_zero.py` checks it against the exact solve for all 2380 states and confirms exactly 514 have a fully flat, all-zero matrix -- the single most common shape on the board.

The standoff



pure equilibrium · $V = +0.000$

Q matrix -- (3, 4, 4, 0)



pure saddle at D/U

Q: PLAYER 0 PAYOFF, OPTIMAL CONTINUATION AFTER THIS TURN

P0 / P1	U	D	L	R
U	0.000000	0.000000	0.000000	0.000000
D	0.000000	0.000000	0.656100	0.000000
L	0.000000	0.000000	0.000000	0.000000
R	0.000000	0.000000	-0.590490	0.000000

ALL 16 SUCCESSOR STATES, FROM THE TRANSITION ENGINE

P0 / P1	U	D	L	R
U	(3, 4, 4, 4, 0)	(3, 4, 4, 3, 0)	(3, 4, 4, 4, 1)	(3, 4, 5, 4, 0)
D	(3, 3, 4, 4, 0)	(3, 3, 4, 3, 0)	(3, 3, 3, 4, 0)	(3, 3, 5, 4, 0)
L	(2, 4, 4, 4, 0)	(2, 4, 4, 3, 0)	(2, 4, 3, 4, 0)	(2, 4, 5, 4, 0)
R	(3, 4, 4, 4, 1)	(4, 4, 4, 3, 0)	(4, 4, 3, 4, 1)	(4, 4, 5, 4, 0)

INDEPENDENT LEARNED CROSS-CHECK: DQN VS. THE EXACT MATRIX

$\max |Q_{\text{DQN}} - Q_{\text{exact}}|$: from zero = **0.3001** · fit to exact = **0.2693**. "From zero" is 400 epochs of TD bootstrap from a random init with no access to the exact answer -- the honest baseline; "fit to exact" is direct supervised regression onto the exact matrices, isolating how well a network this size can even represent Q_{exact} before any bootstrapping noise. One representative run each (seed 0), not the multi-seed study in [docs/neural.md](#). Full per-state numbers for every board, plus a policy-gradient comparison, are in the [interactive explorer's](#) "Show neural cross-check" panel.

P0 / P1	Exact	DQN, from zero	DQN, fit to exact
U/U	0.0000	0.1211	-0.0765
U/D	0.0000	0.0774	0.0149
U/L	0.0000	0.1048	0.0029
U/R	0.0000	0.1415	0.0550
D/U	0.0000	0.2338	0.2693
D/D	0.0000	0.2417	0.1020
D/L	0.6561	0.4552	0.4776
D/R	0.0000	0.2489	-0.0465
L/U	0.0000	0.1602	-0.1495
L/D	0.0000	0.1269	-0.0593
L/L	0.0000	0.1386	-0.0028
L/R	0.0000	0.1168	0.0294
R/U	0.0000	0.2445	0.1532
R/D	0.0000	0.2369	0.1146
R/L	-0.5905	-0.2904	-0.3455
R/R	0.0000	0.1815	-0.0441

4 The open goal

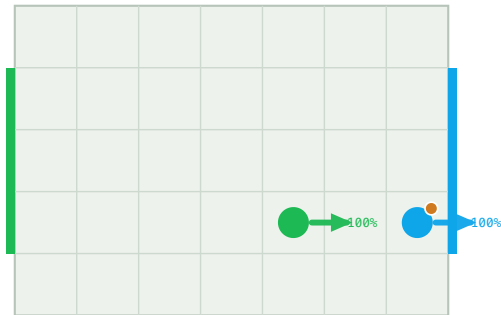
State (6, 1, 4, 1, 0) · player 0 carries the ball · $V = +1.000000$

P0 → R (100%) P1 → R (100%)

Security-tied alternatives (not probabilities): P0 R · P1 U/D/L/R

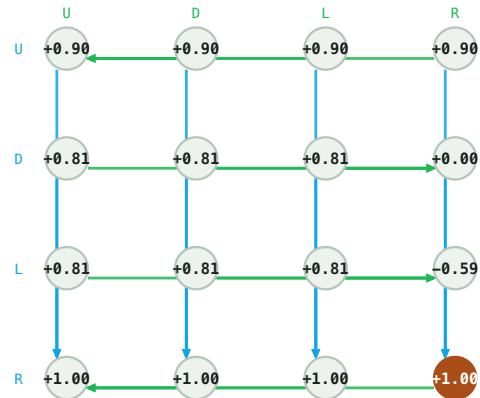
player 0 already sits at the right wall in a goal row, with the ball. Off a goal row this exact same boundary push would just hold in place -- here it scores instead: verified, `game.transitions()` returns a terminal player-0 win for R against every one of player 1's replies. R is the unique optimum (maximin 1.0); player 1's columns all tie at minimax 1.0 too, because there is nothing any of them can do about it.

The open goal



pure equilibrium · $V = +1.000$

Q matrix -- (6, 1, 4, 1, 0)



pure saddle at R/R

Q: PLAYER 0 PAYOFF, OPTIMAL CONTINUATION AFTER THIS TURN

P0 / P1	U	D	L	R
U	0.900000	0.900000	0.900000	0.900000
D	0.810000	0.810000	0.810000	0.000000
L	0.810000	0.810000	0.810000	-0.590490
R	1.000000	1.000000	1.000000	1.000000

ALL 16 SUCCESSOR STATES, FROM THE TRANSITION ENGINE

P0 / P1	U	D	L	R
U	(6, 2, 4, 2, 0)	(6, 2, 4, 0, 0)	(6, 2, 3, 1, 0)	(6, 2, 5, 1, 0)
D	(6, 0, 4, 2, 0)	(6, 0, 4, 0, 0)	(6, 0, 3, 1, 0)	(6, 0, 5, 1, 0)
L	(5, 1, 4, 2, 0)	(5, 1, 4, 0, 0)	(5, 1, 3, 1, 0)	(5, 1, 4, 1, 1)
R	(-1, -1, -1, -1, 0)	(-1, -1, -1, -1, 0)	(-1, -1, -1, -1, 0)	(-1, -1, -1, -1, 0)

INDEPENDENT LEARNED CROSS-CHECK: DQN VS. THE EXACT MATRIX

$\max |Q_{\text{DQN}} - Q_{\text{exact}}|$: from zero = **0.4231** · fit to exact = **0.7772**. "From zero" is 400 epochs of TD bootstrap from a random init with no access to the exact answer -- the honest baseline; "fit to exact" is direct supervised regression onto the exact matrices, isolating how well a network this size can even represent Q_{exact} before any bootstrapping noise. One representative run each (seed 0), not the multi-seed study in [docs/neural.md](#). Full per-state numbers for every board, plus a policy-gradient comparison, are in the [interactive explorer](#)'s "Show neural cross-check" panel.

P0 / P1	Exact	DQN, from zero	DQN, fit to exact
U/U	0.9000	0.9198	1.0031
U/D	0.9000	0.9515	1.0593
U/L	0.9000	0.9224	0.8627

U/R	0.9000	0.8560	0.9428
D/U	0.8100	0.6815	0.7381
D/D	0.8100	0.6681	0.8416
D/L	0.8100	0.7383	0.7128
D/R	0.0000	0.4231	0.5073
L/U	0.8100	0.6379	0.8170
L/D	0.8100	0.7279	0.9228
L/L	0.8100	0.7493	0.7831
L/R	-0.5905	-0.1917	0.1867
R/U	1.0000	1.0035	0.9647
R/D	1.0000	0.9963	1.0290
R/L	1.0000	1.1082	1.2321
R/R	1.0000	0.9655	0.9798

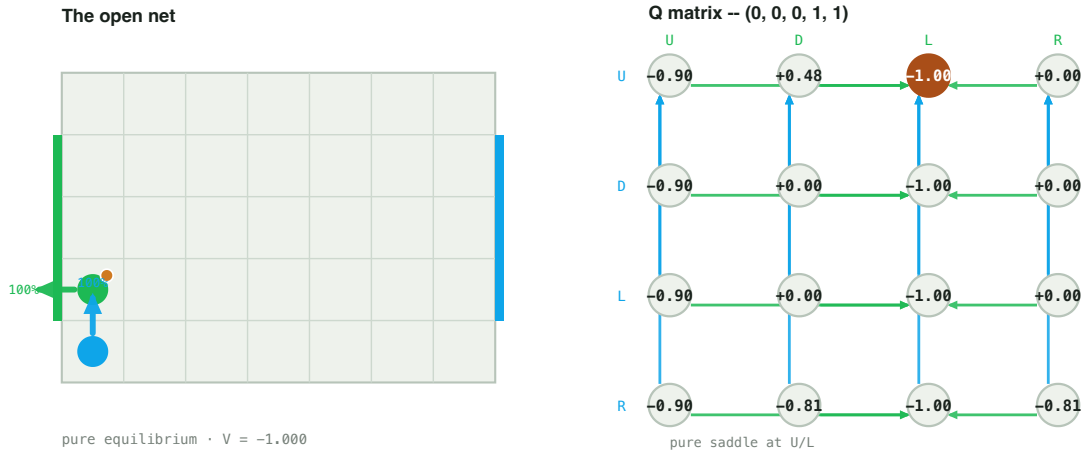
5 The open net

State (0, 0, 0, 1, 1) · player 1 carries the ball · $V = -1.000000$

P0 → U (100%) P1 → L (100%)

Security-tied alternatives (not probabilities): P0 U/D/L/R · P1 L

the mirror of the open goal from player 1's side: player 1 is at the left wall in a goal row, ball in hand, and L scores unconditionally (verified against every reply). The bleaker half of this case is player 0's row: every one of U/D/L/R ties at the exact same -1.0 -- there is nothing player 0 can do differently that changes the outcome, which is itself worth seeing on a board rather than only in a number.



Q: PLAYER 0 PAYOFF, OPTIMAL CONTINUATION AFTER THIS TURN

P0 / P1	U	D	L	R
U	-0.900000	0.478297	-1.000000	0.000000
D	-0.900000	0.000000	-1.000000	0.000000
L	-0.900000	0.000000	-1.000000	0.000000
R	-0.900000	-0.810000	-1.000000	-0.810000

ALL 16 SUCCESSOR STATES, FROM THE TRANSITION ENGINE

P0 / P1	U	D	L	R
U	(0, 1, 0, 2, 1)	(0, 1, 0, 0, 0)	(-1, -1, -1, -1, 1)	(0, 1, 1, 1, 1)
D	(0, 0, 0, 2, 1)	(0, 0, 0, 1, 0)	(-1, -1, -1, -1, 1)	(0, 0, 1, 1, 1)
L	(0, 0, 0, 2, 1)	(0, 0, 0, 1, 0)	(-1, -1, -1, -1, 1)	(0, 0, 1, 1, 1)
R	(1, 0, 0, 2, 1)	(1, 0, 0, 0, 1)	(-1, -1, -1, -1, 1)	(1, 0, 1, 1, 1)

INDEPENDENT LEARNED CROSS-CHECK: DQN VS. THE EXACT MATRIX

$\max |Q_{\text{DQN}} - Q_{\text{exact}}|$: from zero = **0.4947** · fit to exact = **0.4049**. "From zero" is 400 epochs of TD bootstrap from a random init with no access to the exact answer -- the honest baseline; "fit to exact" is direct supervised regression onto the exact matrices, isolating how well a network this size can even represent Q_{exact} before any bootstrapping noise. One representative run each (seed 0), not the multi-seed study in [docs/neural.md](#). Full per-state numbers for every board, plus a policy-gradient comparison, are in the [interactive explorer](#)'s "Show neural cross-check" panel.

P0 / P1	Exact	DQN, from zero	DQN, fit to exact
U/U	-0.9000	-0.6967	-0.6653
U/D	0.4783	0.3357	0.3346
U/L	-1.0000	-0.7185	-0.7320

U/R	0.0000	-0.3544	0.0854
D/U	-0.9000	-0.9347	-1.0265
D/D	0.0000	-0.1585	-0.3353
D/L	-1.0000	-0.9130	-0.8393
D/R	0.0000	-0.3615	-0.4049
L/U	-0.9000	-0.7078	-0.7625
L/D	0.0000	-0.0688	0.1335
L/L	-1.0000	-0.8956	-0.7354
L/R	0.0000	-0.2438	-0.0471
R/U	-0.9000	-0.9099	-0.9654
R/D	-0.8100	-0.3153	-0.5793
R/L	-1.0000	-1.0282	-0.8748
R/R	-0.8100	-0.5346	-0.5617

6 The getaway

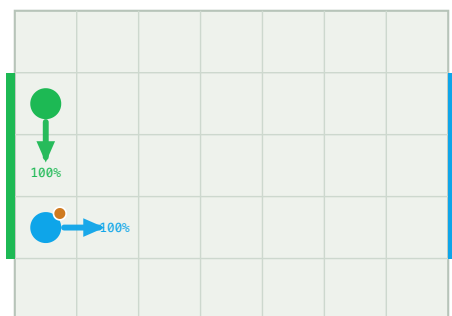
State (0, 1, 0, 3, 0) · player 0 carries the ball · $V = +0.531441$

P0 → R (100%) P1 → D (100%)

Security-tied alternatives (not probabilities): P0 R · P1 U/D/L/R

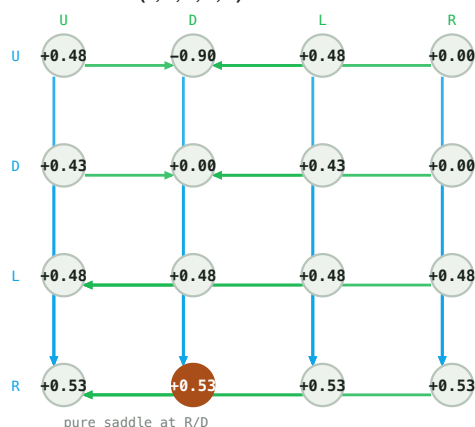
R is player 0's *unique* safe action (0.531441, no tie) -- an entirely independent move to (1,1) that never depends on player 1's reply. U looks just as reasonable but is a trap: against D, both target the same empty cell (0,2); player 0 wins the race there (verified: (U, D) → (0,2,0,3,1)) and immediately loses the ball anyway, crashing to -0.9. Player 1's columns all tie at 0.531441 -- there is no reply that slows the escape down.

The getaway



pure equilibrium · $V = +0.531$

Q matrix -- (0, 1, 0, 3, 0)



Q: PLAYER 0 PAYOFF, OPTIMAL CONTINUATION AFTER THIS TURN

P0 / P1	U	D	L	R
U	0.478297	-0.900000	0.478297	0.000000
D	0.430467	0.000000	0.430467	0.000000
L	0.478297	0.478297	0.478297	0.478297
R	0.531441	0.531441	0.531441	0.531441

ALL 16 SUCCESSOR STATES, FROM THE TRANSITION ENGINE

P0 / P1	U	D	L	R
U	(0, 2, 0, 4, 0)	(0, 2, 0, 3, 1)	(0, 2, 0, 3, 0)	(0, 2, 1, 3, 0)
D	(0, 0, 0, 4, 0)	(0, 0, 0, 2, 0)	(0, 0, 0, 3, 0)	(0, 0, 1, 3, 0)
L	(0, 1, 0, 4, 0)	(0, 1, 0, 2, 0)	(0, 1, 0, 3, 0)	(0, 1, 1, 3, 0)
R	(1, 1, 0, 4, 0)	(1, 1, 0, 2, 0)	(1, 1, 0, 3, 0)	(1, 1, 1, 3, 0)

INDEPENDENT LEARNED CROSS-CHECK: DQN VS. THE EXACT MATRIX

$\max |Q_{\text{DQN}} - Q_{\text{exact}}|$: from zero = **0.6241** · fit to exact = **0.6223**. "From zero" is 400 epochs of TD bootstrap from a random init with no access to the exact answer -- the honest baseline; "fit to exact" is direct supervised regression onto the exact matrices, isolating how well a network this size can even represent Q_{exact} before any bootstrapping noise. One representative run each (seed 0), not the multi-seed study in [docs/neural.md](#). Full per-state numbers for every board, plus a policy-gradient comparison, are in the [interactive explorer](#)'s "Show neural cross-check" panel.

P0 / P1	Exact	DQN, from zero	DQN, fit to exact
U/U	0.4783	0.2979	0.5357
U/D	-0.9000	-0.2759	-0.2777
U/L	0.4783	0.2296	0.3890

U/R	0.0000	0.0898	0.1222
D/U	0.4305	0.3734	0.4584
D/D	0.0000	0.2029	0.0880
D/L	0.4305	0.2627	0.3620
D/R	0.0000	0.1858	0.1598
L/U	0.4783	0.3046	0.4892
L/D	0.4783	-0.0476	-0.0927
L/L	0.4783	0.3345	0.4042
L/R	0.4783	0.1470	0.2213
R/U	0.5314	0.4519	0.5313
R/D	0.5314	0.4052	0.4881
R/L	0.5314	0.4103	0.4914
R/R	0.5314	0.4345	0.4883

7 The tightest tie

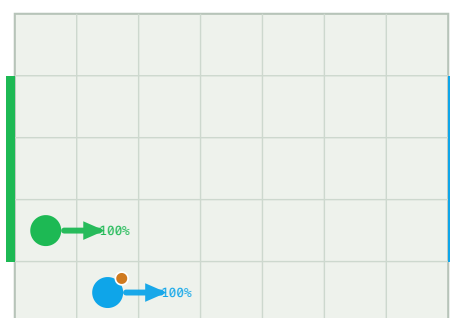
State (1, 0, 0, 1, 0) · player 0 carries the ball · $V = +0.000000$

P0 → R (100%) P1 → R (100%)

Security-tied alternatives (not probabilities): P0 D/R · P1 R

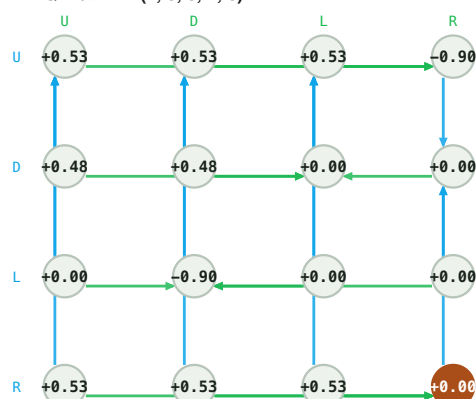
no state on this entire board ever has a fully unique best action for *both* players simultaneously -- checked exhaustively, (1,1) never occurs among all 2380 states. This is the smallest tie shape that does occur (only 16 of 2380 states): player 0's D and R tie exactly; player 1's R is uniquely optimal. U and L each look just as safe as D/R, but each hides its own contest: (U, R) both target the same cell (1,1), and (L, D) both target (0,0) -- verified below, the carrier wins the square but the ball still goes to the defender either way -- while D and R never target the same cell as player 1 at all.

The tightest tie



pure equilibrium · $V = +0.000$

Q matrix -- (1, 0, 0, 1, 0)



pure saddle at R/R

Q: PLAYER 0 PAYOFF, OPTIMAL CONTINUATION AFTER THIS TURN

P0 / P1	U	D	L	R
U	0.531441	0.531441	0.531441	-0.900000
D	0.478297	0.478297	0.000000	0.000000
L	0.000000	-0.900000	0.000000	0.000000
R	0.531441	0.531441	0.531441	0.000000

ALL 16 SUCCESSOR STATES, FROM THE TRANSITION ENGINE

P0 / P1	U	D	L	R
U	(1, 1, 0, 2, 0)	(1, 1, 0, 0, 0)	(1, 1, 0, 1, 0)	(1, 1, 0, 1, 1)
D	(1, 0, 0, 2, 0)	(1, 0, 0, 0, 0)	(1, 0, 0, 1, 0)	(1, 0, 1, 1, 0)
L	(0, 0, 0, 2, 0)	(0, 0, 0, 1, 1)	(0, 0, 0, 1, 0)	(0, 0, 1, 1, 0)
R	(2, 0, 0, 2, 0)	(2, 0, 0, 0, 0)	(2, 0, 0, 1, 0)	(2, 0, 1, 1, 0)

INDEPENDENT LEARNED CROSS-CHECK: DQN VS. THE EXACT MATRIX

$\max |Q_{\text{DQN}} - Q_{\text{exact}}|$: from zero = **0.3929** · fit to exact = **0.5398**. "From zero" is 400 epochs of TD bootstrap from a random init with no access to the exact answer -- the honest baseline; "fit to exact" is direct supervised regression onto the exact matrices, isolating how well a network this size can even represent Q_{exact} before any bootstrapping noise. One representative run each (seed 0), not the multi-seed study in [docs/neural.md](#). Full per-state numbers for every board, plus a policy-gradient comparison, are in the [interactive explorer](#)'s "Show neural cross-check" panel.

P0 / P1	Exact	DQN, from zero	DQN, fit to exact
U/U	0.5314	0.3744	0.4738

U/D	0.5314	0.3351	0.1921
U/L	0.5314	0.2442	0.5397
U/R	-0.9000	-0.5770	-0.3602
D/U	0.4783	0.2666	0.2314
D/D	0.4783	0.0854	0.2215
D/L	0.0000	0.2194	0.4356
D/R	0.0000	0.0576	-0.2646
L/U	0.0000	0.1828	0.0154
L/D	-0.9000	-0.7350	-0.5569
L/L	0.0000	-0.0909	0.1429
L/R	0.0000	-0.0770	-0.2302
R/U	0.5314	0.4398	0.5306
R/D	0.5314	0.4147	0.5372
R/L	0.5314	0.4231	0.5675
R/R	0.0000	0.2691	0.1459

8 The dead zone

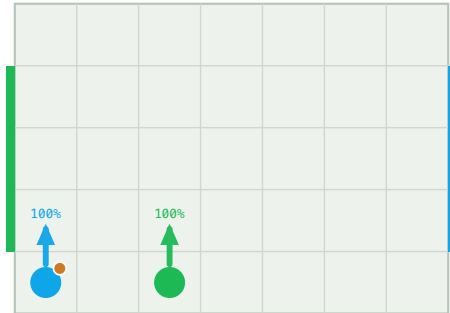
State (0, 0, 2, 0, 0) · player 0 carries the ball · $V = +0.000000$

P0 → U (100%) P1 → U (100%)

Security-tied alternatives (not probabilities): P0 U/D/L/R · P1 U/D/L/R

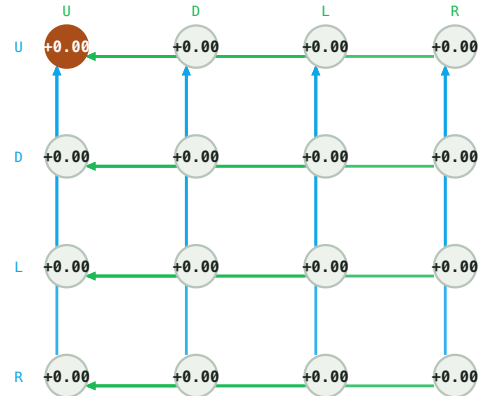
the single most common shape on the whole board (514 of 2380 states): both players are fully indifferent among all four actions. It is not merely that the guaranteed values tie -- every one of the 16 cells in the Q matrix is exactly 0.0, including a genuine ball-swapping contest ((R, L) → a state that is itself also worth exactly 0, verified below). All successors have the same optimal-continuation value, although their physical positions differ.

The dead zone



pure equilibrium · $V = +0.000$

Q matrix -- (0, 0, 2, 0, 0)



pure saddle at U/U

Q: PLAYER 0 PAYOFF, OPTIMAL CONTINUATION AFTER THIS TURN

P0 / P1	U	D	L	R
U	0.000000	0.000000	0.000000	0.000000
D	0.000000	0.000000	0.000000	0.000000
L	0.000000	0.000000	0.000000	0.000000
R	0.000000	0.000000	0.000000	0.000000

ALL 16 SUCCESSOR STATES, FROM THE TRANSITION ENGINE

P0 / P1	U	D	L	R
U	(0, 1, 2, 1, 0)	(0, 1, 2, 0, 0)	(0, 1, 1, 0, 0)	(0, 1, 3, 0, 0)
D	(0, 0, 2, 1, 0)	(0, 0, 2, 0, 0)	(0, 0, 1, 0, 0)	(0, 0, 3, 0, 0)
L	(0, 0, 2, 1, 0)	(0, 0, 2, 0, 0)	(0, 0, 1, 0, 0)	(0, 0, 3, 0, 0)
R	(1, 0, 2, 1, 0)	(1, 0, 2, 0, 0)	(1, 0, 2, 0, 1)	(1, 0, 3, 0, 0)

INDEPENDENT LEARNED CROSS-CHECK: DQN VS. THE EXACT MATRIX

$\max |Q_{\text{DQN}} - Q_{\text{exact}}|$: from zero = **0.1744** · fit to exact = **0.1498**. "From zero" is 400 epochs of TD bootstrap from a random init with no access to the exact answer -- the honest baseline; "fit to exact" is direct supervised regression onto the exact matrices, isolating how well a network this size can even represent Q_{exact} before any bootstrapping noise. One representative run each (seed 0), not the multi-seed study in [docs/neural.md](#). Full per-state numbers for every board, plus a policy-gradient comparison, are in the [interactive explorer](#)'s "Show neural cross-check" panel.

P0 / P1	Exact	DQN, from zero	DQN, fit to exact
U/U	0.0000	0.0451	-0.0391
U/D	0.0000	0.0795	-0.0000
U/L	0.0000	0.0911	0.1498

U/R	0.0000	0.0089	0.0374
D/U	0.0000	0.0130	0.0521
D/D	0.0000	0.0878	-0.0250
D/L	0.0000	-0.0387	0.0207
D/R	0.0000	0.0907	0.0729
L/U	0.0000	0.0408	0.0076
L/D	0.0000	0.0560	-0.0179
L/L	0.0000	-0.0322	-0.0095
L/R	0.0000	0.0744	0.0280
R/U	0.0000	0.0784	-0.0684
R/D	0.0000	0.0770	0.0203
R/L	0.0000	-0.1744	-0.0493
R/R	0.0000	0.0457	-0.0414